

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

QIAN QI

ABSTRACT. The weighted cotangent dual is the vector space of invariant signed Radon measures; invariant probabilities form its positive normalized slice. Equality of the full perturbed pressure functional is treated platform by platform. Transfer jets are typed on regularity levels, while physical memory uses a separate common-domain Hilbert form bundle. Optional projections and source versus state contraction derivatives are then rigorously separated.

Platform identifier: TYPED-COPRODUCT-v4. **Controlling revision:** ROUND12-REFEREE-POSITIVE-CLOSURE. **Registered proof source:** C2_SIGNED_DUAL_FORM_MEMORY.tex.

CONTENTS

1. Round-twelve positive closure: signed cotangents, form bundles, and covariant memory	1
1.1. The weighted invariant quotient	2
1.2. Full-pressure rigidity, platform by platform	2
1.3. Transfer jets and a separate Hilbert form bundle	3
1.4. Likelihoods and contractions	3
Scope and verification status	4

1. ROUND-TWELVE POSITIVE CLOSURE: SIGNED COTANGENTS, FORM BUNDLES, AND COVARIANT MEMORY

Invariant-integral equivalence, equality of one pressure number, and equality of the full perturbed pressure functional are different notions. The continuous dual is a vector space of signed invariant Radon measures; probabilities are a normalized positive slice.

Date: August 31, 2026.

2020 *Mathematics Subject Classification.* Primary 60F10, 60H10; Secondary 37A50, 49L25.

Key words and phrases. path-space large deviations, cotangent fields, entropic rigidity, tangent laws, Girsanov, BSDE.

1.1. The weighted invariant quotient. Let $\mathcal{A}_W$ be the weighted strict potential space and

$$\mathcal{N}_{\text{inv}} = \overline{\text{span}}\{U - U \circ \Theta_t : t \in \mathbb{Q}\}.$$

Constants are kept separate.

Theorem 1.1 (Signed invariant duality). *Assume the platform pointwise Lyapunov estimate*

$$\frac{1}{T} \int_0^T W(\Theta_t x) dt \leq C(1 + W(x))$$

for all x and $T \geq 1$. Then

$$(\mathcal{A}_W / \mathcal{N}_{\text{inv}})' = \{\text{finite invariant signed Radon measures with finite } W\text{-moment}\}.$$

Invariant probabilities are exactly the positive normalized slice. Two potentials have equal integrals under every invariant probability iff their difference lies in $\mathbb{R}1 + \mathcal{N}_{\text{inv}}$.

Proof. The weighted strict dual is the signed weighted Radon space. A quotient functional is therefore a signed measure annihilating every rational-time coboundary, hence invariant. Conversely invariance annihilates the closed span. For separation, average a signed separator in time. The pointwise Lyapunov estimate gives uniform weighted tightness after integration against its total variation; every weak limit is invariant and preserves the pairing. Jordan components of an invariant signed measure are invariant by uniqueness of the Jordan decomposition. Normalizing a nonzero positive component gives the probability separator. $\square$

1.2. Full-pressure rigidity, platform by platform. For a base potential F write $\mathcal{P}_F(H) = P(F + H) - P(F)$.

Theorem 1.2 (Perturbed-pressure rigidity). *On the Sinai Hölder platform, $\mathcal{P}_F(H) = \mathcal{P}_G(H)$ for every small Hölder H iff $F - G$ is a constant plus a Hölder temporal coboundary. On the regular hard-sphere source platform the same identity holds iff the source difference belongs to the closed $B3$ balance/endpoint covariance gauge plus a constant. Equality of one scalar pressure value is not used.*

Proof. On the Sinai platform, equality of all perturbations gives equality of every perturbed equilibrium law. Periodic approximation then gives equal sums of $F - G$ on all periodic orbits; the standard stable/unstable telescoping construction solves the Hölder cohomological equation. On the hard-sphere platform, differentiating all finite source perturbations gives equality of the complete mean and covariance functionals. B3's exact local-noise representation identifies the annihilator with the closed balance/endpoint gauge. Constants account for the pressure normalization. $\square$

1.3. Transfer jets and a separate Hilbert form bundle. The finite Riesz range of the Sinai transfer operator is trivialized by a fixed family $e_i(\eta) = \Pi_\eta v_i$ on each chart. Its q th derivative is viewed as a map from strong level $m + q$ to level m ; no infinite Picard transport is used.

For memory, let $\mathcal{H}_\eta = L^2(\widehat{\pi}_\eta)$ and let $\mathfrak{a}_\eta$ be the closed Doob generator form on a common Gelfand domain $\mathcal{V}$. Pull the inner products to one reference Hilbert space by the positive square root of the correlation Gram operator.

Theorem 1.3 (Common-domain form bundle). *On a regular phase chart, $\mathfrak{a}_\eta$ is a type-(B) differentiable family of closed sectorial forms on the common domain $\mathcal{V}$. Consequently the generators have norm-resolvent derivatives*

$$D(z - L_\eta)^{-1} = (z - L_\eta)^{-1}(DL_\eta)(z - L_\eta)^{-1}$$

in the graph-form sense. After quotienting constant/conservation null modes, the Gram matrix of a fixed finite resolved observable family has a uniform positive lower eigenvalue and defines a differentiable orthogonal projection R_η .

Proof. A2/A4 source derivatives of correlations are bounded on fresh strong levels; their pullbacks define bounded form derivatives on $\mathcal{V}$. Uniform sector bounds and Kato's form theorem give the common-domain resolvent family. The finite correlation Gram matrix is continuous; remove its exact nullspace at the base point and shrink the chart so the remaining eigenvalues stay positive. The usual finite-matrix formula then differentiates R_η . $\square$

Theorem 1.4 (Covariant compressed memory). *Away from the finitely many tracked transmission zeros, the compressed memory*

$$\widehat{K}_\eta(z) = zR_\eta - R_\eta L_\eta R_\eta - [R_\eta(z - L_\eta)^{-1}R_\eta]^{-1}$$

is differentiable in source. Its derivative includes the form derivative of L_η , both derivatives of R_η , the correlation pairing derivative, and the derivative of every descriptor residue. The formula is independent of the finite transfer-eigenline frame.

Proof. Theorem 1.3 supplies a common resolvent domain and DR_η . Differentiate the finite compressed matrix and use $D(C^{-1}) = -C^{-1}(DC)C^{-1}$. A4's normalized meromorphic decomposition tracks zeros and residues; differentiation of their Riesz contours gives the descriptor terms. The transfer frame enters only the proof of correlation response, not the Hilbert-space identification. $\square$

1.4. Likelihoods and contractions.

Theorem 1.5 (Optional projection and diffusion likelihood). *Under the A4 uniform quenched kernel theorem and the common exponential moment bound, finite-history likelihood martingales and their optional projections converge stably to the Brownian stochastic exponential. In the limiting Brownian filtration the martingale representation theorem gives the Girsanov drift and the unique bounded quadratic BSDE with the declared terminal data.*

Proof. Conditional-kernel convergence identifies finite-cylinder conditional expectations. Uniform $L^{1+\delta}$ bounds give uniform integrability and permit passage from cylinders to the complete filtration. Joint rough convergence identifies the limiting martingale integrand. Brownian martingale representation gives the stochastic exponential; exponential transformation of the bounded terminal value gives the unique quadratic BSDE. $\square$

Let $C : X \rightarrow Y$ be a differentiable state contraction. Source and state derivatives are written separately.

Proposition 1.6 (Typed contraction derivatives). *For a source perturbation $h : Y \rightarrow \mathbb{R}$,*

$$DQ_C(\varphi)[h] = DQ(\varphi \circ C)[h \circ C].$$

For a state tangent $v \in T_x X$, $D(\varphi \circ C)(x)v = D\varphi(Cx)DC(x)v$. The corresponding Hessian contains the ordinary second state derivative of C ; it is not a source Hessian identity.

Proof. The first identity follows by differentiating the exact source composition. The second and its Hessian are the ordinary Banach-space chain rule. $\square$

Theorem 1.7 (Round-twelve C2 closure). *The cotangent dual is the invariant signed-measure space, full-pressure rigidity is platform-specific, transfer regularity loss is explicitly typed, the physical memory bundle has a common form domain, and likelihood and contraction derivatives are correctly separated.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.4, and 1.5, with Proposition 1.6. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn